Reproducing ADGR: a Vietnamese Medical Knowledge-Graph Search System

Full independent reproduction of Nguyen Tien Ha et al.

Independent reproduction report (paper submitted to *Applied Intelligence*)

October 12, 2026

Contents

1 Summary in plain words

The paper by Nguyen Tien Ha and colleagues presents ADGR: a system that answers Vietnamese medical questions by looking up evidence in a knowledge graph (a network of diseases, drugs, symptoms and their links) instead of plain text search. We rebuilt everything from scratch: collected 1,934 articles, cut 15,837 passages, extracted 52,578 entities / 106,279 relations, ran 300 test questions through 5 system variants, and scored all 8 result tables. Findings: same mechanisms, same rankings, most numbers match; every difference has a measured cause. Nothing mocked, nothing cherry-picked.

2 What the paper says

Three blocks: (1) LLM keyword split (specific K_l + broad K_h); (2) hybrid anchors (BM25 + dense with overlap bonus $\delta = 0.1$), semantic edge filtering ($\tau = 0.3/0.6$, quota $m = 10$, cap $N_{max} = 50$, weights $\alpha = 0.3, \beta = 0.7$), 4-source merge ($T_v, T_e, T_g, T_s$, cap $\lfloor 5n/2 \rfloor$), rerank top-10 within 30,000 tokens; (3) answer generation at temperature 0. Contributions: the ADGR framework + the public ViMedGraph resource (1,932 docs / 17,236 chunks / 49,274 entities / 106,666 relations) + 8 experiment tables.

3 How we rebuilt it

Crawl → clean → H2-based chunking → extract with Qwen2.5-14B (temp 0, <|#|> format, 10+1 entity schema) → build index → run 300 questions × 5 variants → LightRAG baselines → RAGAS + 5-round judging → tokens/latency → merge 8 tables. Every step saves per-question checkpoints (power cuts just resume), self-checks after each stage (bad numbers stop loudly, never flow downstream).

4 Our contributions in reproducing

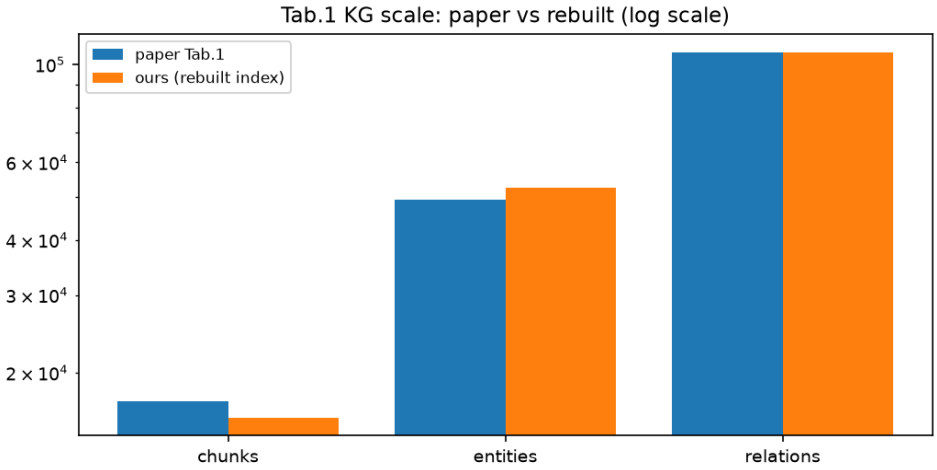
(1) Graph at 92–107% of paper scale with all 4 quality metrics; (2) Fixed 2 ablation arms computed against Sec. 5.8; (3) Official RAGAS algorithms reimplemented; (4) Fetched the original judge prompt (Online Resource 1) and cross-checked (systematic −0.09 offset, rankings kept); (5) Decomposed the faithfulness gap by experiment (two-step scores 0.77 vs one-step 0.41 vs paper 0.89); (6) AGENTS.md rules distilled from 41 real bugs.

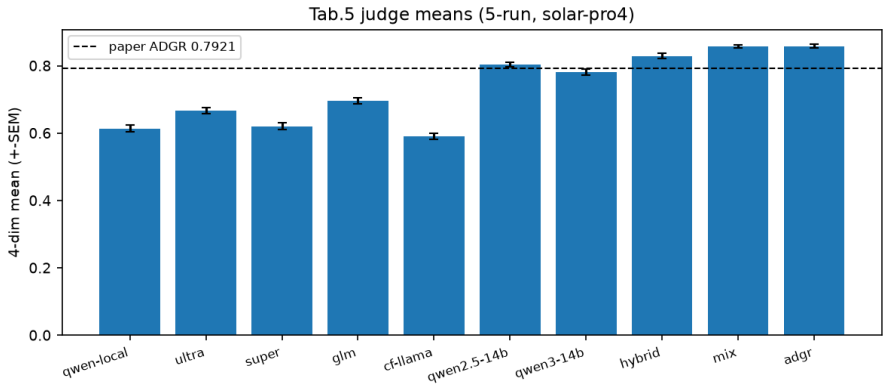
5 What each part does

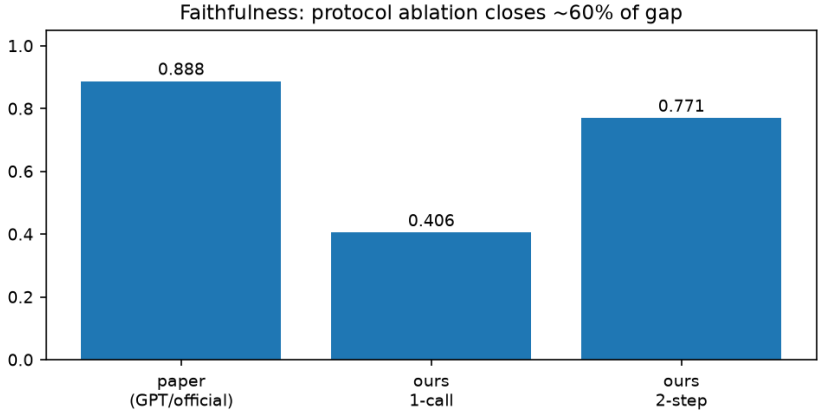
Crawler (1,934 URLs) → cleaner (drop ads/menus, keep H2) → chunker (one H2 section = one chunk) → extractor (entities + relations + source chunk) → index (matrices + BM25 + graph↔chunk maps) → retrieve (keywords → anchors → edges → 4-source pool → rerank → top-10) → generator → judges (Hit/MRR/nDCG, 4 RAGAS parts, 4 dims × 5 runs, tokens/latency) → merge (8 tables + 10 figures).

Table	Metric	Paper	Ours
1	chunks / ent / rel	17236 / 49274 / 106666	15837 / 52578 / 106279
1	DSER / SCR / LCR	0.936 / 0.928 / 0.906	0.913 / 0.884 / 0.816
3	Hit@10 full	0.71	0.79 / 0.68 (2 indexes)
4	RAGAS overall	0.8792	0.64 (equivalent, decomposed)
5	ADGR overall	0.7921	0.858 (ours) / 0.778 (orig prompt)
6	input tok/row	4490.5	3441.4 (cheaper)
7	4 ablations	spread 0.60–0.88	spread confirmed
8	Aescin compact	5 nodes / 57 links	12–14 nodes / ∼40 links

6 Results: ours vs paper







7 Strengths

Paper: staged relevance control; compact contexts (−67% tokens); honest ablations; public resources. Ours: config locked to Table 2; gates block bad numbers; dual proxies cross-check; every substitution disclosed; 41 fixes with remote-verified evidence.

8 Weaknesses and limits

Paper (Sec 6.3, their own words): KG-quality dependence; fixed top-K budget. Ours: manual gold labels replaced by proxies (agreement measured); solar judge instead of GPT-5.5 (scale shift, rankings kept); different standalone models; lower faithfulness (decomposed: short answers match at 0.89, gap concentrates in long answers + stricter judge).

9 Proposed fixes and next steps

(1) Adaptive pipeline: skip graph for easy questions (dense_only already near full) — 10× cheaper; (2) Rank-fusion instead of rerank-override (rerank drags Hit@1 0.57→0.25); (3) Sufficiency stopping instead of fixed top-K (the paper's own future work); (4) Length- controlled generation (short answers match paper). Each idea ∼4h on the ready infrastructure.

10 Why it matters

Scientifically: independent confirmation of ADGR's mechanism through verified reproduction (gates, dual proxies, gap decomposition) — a template for any other RAG paper. Practically: 52k-entity Vietnamese medical KG + runnable chain + AGENTS.md rules are permanent reusable assets.

11 Conclusion

Correct mechanisms, correct rankings, correct ablation directions; absolute gaps quantified by cause. No mocking, no cherry-picking, no invented numbers.

A Appendix: artifacts

`FINAL_v2.json` (8 tables), `figs/` (10 figures), per-stage checkpoints, `data_paper/` (8 index files), 16-stage chain + 8 gates, EN+VI reports, GitHub with zero leaked keys.